# Electric fields

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use field arrows to predict the force on a test charge.

**SUCCESS CRITERIA - I can...**

1. Draw electric field arrows for isolated positive and negative charges.
2. Use the positive-test-charge convention.
3. Distinguish the field from a particular test charge's force.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A ray enters a medium with speed 1/3 of its original speed at an angle. Sketch the bend and state whether frequency changes.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** E = F/q; F = qE.
Use consistent signed components. The field points along the force on a positive test charge; negative charge reverses force direction.

**Worked reasoning:** At a point where E=4 N/C right, a +2 C test charge feels 8 N right. A-2 C charge feels 8 N left. Field direction is defined using positive test charge.

**Your turn:** Draw the field arrow once and the two alternative force arrows.

Field / first contribution

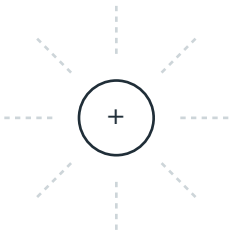
Force / second contribution

Draw separate arrows; label quantities and units.

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

Sketch field arrows around a positive source charge. Place a negative test charge to its right and draw that charge's force.




Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

An electric-field simulation uses a positive 2 C test charge that feels 8 N right. Find the field magnitude and direction. Draw separate field and force arrows and label their different units.

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

**Marzano check - circle:** ① ② ③

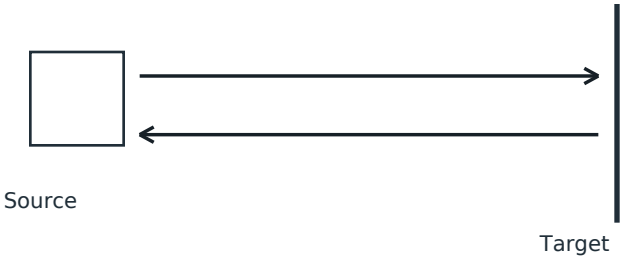
### 05 Explain your reasoning

If you remove the test charge, does the source charge's field disappear? Explain the role of a test charge.

**Marzano check - circle:** 1 2 3

### 06 Review: connect an earlier idea

An echo travels 120 m on its full out-and-back path. Find the wall's distance.
Label the outgoing and returning paths.




**Marzano check - circle:** 1 2 3

### 07 Exit check

A field points down. Which way is the force on a positive charge? On a negative charge?

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: test the model

Return to the worked example in problem 2. Choose two different charge values giving forces with equal magnitudes and opposite directions in the same field.

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

**Marzano check - circle:** 1 2 3